University of Toronto

CSC490

A1

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1 Interest Statements

We want to work on the problem of labelling brand presence in Formula 1 (F1) race videos. During any entertainment event, there exists a symbiotic relationship between the event and brands: brands pay to have their logos on display, their items used, or commercials played and events get money to make the event better. Brands do this based on whether the event has proven to receive viewership and their brand can access the audience's eyes. However, the majority of viewers for events are seen on a screen compared to in person, so it is unclear whether brands do receive an appropriate screen time for their investment. Does the higher paying brand receive more or less screen time than one that pays less? We aim to answer this question for F1, a growing sport with iconic branding located in positions meant for the online viewer (e.g. the top of the car where a camera is positioned above the driver). As aspiring computer scientists with relevant ML experience and a passion for F1, we are taking the opportunity to combine our technical interests with a sport we like, while addressing real inefficiencies in sports marketing. With AI/ML being used to automate more and more tasks, we believe this one can be solved using machine learning and computer vision, innovating fields we are particularly interested in gaining skills for. Specifically, this problem falls in the broad technical domain of video segmentation and categorization and will likely require supervised fine-tuning from existing vision models. Ideally, our solution will be able to label videos in real-time, and this poses several infrastructure requirements. The technical specifications will be in section 3 (project outline) and the business problem we're solving will be in outlined in section 4 (press release).

Excited to contribute

- Codi: I am excited to work on optimizing model inference to predict at (pseudo) real-time speeds. For example, we might need to play with different approaches to batch frame inputs, or space the frame input to the model.
- Yusuf: I'm looking forward to working on making our video models run faster and lighter through techniques learnt in class so we can label brand presence in F1 races in real time.
- Mikhail: I am excited to apply and build upon my current skills during the development of this project. Specifically, I am interested in building the infrastructure for the project as that isn't something I have done before.
- Rudraksh: I'm excited to work towards making the logo detection pipeline more robust to visual perturbations such as blur, noise, crop, rotations, scale, etc. - which are extremely common in F1 videos/streams.

2 Landscape analysis

The landscape analysis shows that while many commercial platforms (e.g., Relo Metrics, Blinkfire, Sponsorlytix) already measure brand exposure, most are proprietary and not tailored specifically for the unique challenges of Formula 1, such as high-speed motion, small and obstructed logos, and complex camera angles. Research validates the use of YOLO-based detection and segmentation approaches but has not yet been specialized to motorsports. We have a clear opportunity for our project to bridge these gaps with an open, adaptable, and F1-specific approach. By combining ideas from commercial systems and research prototypes, our solution aims to deliver real-time sponsor detection that can serve as a foundation for practical use in sports analytics. See the full summary table in **Appendix A**, Table 1.

3 Project outline

3.1 Problem statement

We aim to build a system that detects and measures the on-screen presence of sponsor brands in Formula 1 race videos in near real time. The challenge lies in robust logo detection under high-speed motion, blur and frequent camera cuts. Current commercial tools are proprietary and not specialized for F1, creating an opportunity for an open and adaptable solution. With access to a dataset of annotated F1 video frames, we are well-positioned to fine-tune a state-of-the-art object detector. Our system will output per-frame detections, track logos across time, and aggregate them into brand exposure metrics that are meaningful for sponsors and teams.

3.2 Proposed solution

We propose a detection-first pipeline centered on a fine-tuned YOLO model trained on our F1 logo dataset. During training, frames are extracted and augmented (scale, blur, brightness, perspective), annotated with bounding boxes, and used to fine-tune a pretrained YOLO backbone. At inference, race footage is decoded, preprocessed, and passed through the YOLO model to output bounding boxes and brand classifications. These are linked across frames using a lightweight tracker (ByteTrack), enabling the continuity of brand IDs and the measurement of dwell time. The system will generate exposure statistics (seconds on screen, relative logo area, share of voice) per brand and visualize them in a simple front-end dashboard. To achieve real-time performance, we will explore quantization and perhaps CUDA-accelerated inference.

One point that we need to consider is how to handle new logos that may appear in following Formula One seasons. An approach to overcome this could be to find a closed set of sponsors per season and fine-tune our model once per year as a maintenance process. A more robust but potentially more complex and inaccurate approach would be to have a two-step architecture. The first step would be to use a pre-trained model for detecting logo's. The second step would be to use a brand classifier head like a CNN/ViT head trained on logo crops.

3.3 Technical milestones

Our project milestones begin with preparing the dataset by cleaning and augmenting the existing F1 logo frames to simulate race conditions. We will then budget compute resources to set realistic throughput targets, followed by building a pipeline skeleton for real-time video ingestion and mock model integration. On the modeling side, we will select and fine-tune a YOLO variant for brand detection, validate it on held-out frames, and then optimize inference using CUDA acceleration and quantization. Next, we will integrate tracking (e.g., ByteTrack) to maintain logo continuity across frames and compute exposure metrics such as dwell time, area, and share-of-voice. A lightweight dashboard will visualize detections and exposure charts in real time, with an extended ROI module estimating sponsorship value by weighting exposure with prominence and viewership. Finally, we will stress-test the full system on race-length footage and present our findings in the demo.

3.4 Unknowns

Several uncertainties remain. It is unclear how well the dataset will generalize to varied lighting, angles, and broadcast styles, or whether augmentation alone will suffice. We must evaluate the trade-offs between fine-tuning and zero-shot approaches for unseen brands, as well as whether separating types of visibility (on-car, trackside, virtual) adds business value. Real-time constraints also pose open questions: what maximum latency is acceptable, how frequently frames should be processed, and whether batching or multi-frame context is needed. Finally, we need to assess tolerance for inaccuracy, model robustness to logo variations and distortions, and whether preprocessing (e.g., denoising or normalization) improves performance.

4 Project Press release

Visight

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Visight offers a precise, state-of-the-art measurement of brand exposure in F1 broadcasting

AI-powered video analysis provides accurate metrics for brand sponsorships displayed in broadcasts

Today, Visight is proud to announce the launch of a groundbreaking sports sponsorship analytics platform that uses computer vision to analyze sports broadcasts and provide brand exposure measurements. This product allows brands to determine their screen time during events they have partnered to determine the exposure value they are receiving for their investment. The data-driven insights from this product allow brands, agencies, and sports properties to optimize their sponsorship investments and determine where to get the best bang for their buck.

PwC estimates the sports sponsorship market will grow to $109.1 billion by 2030 (source), and in 2024 Sportico reported that F1 teams generated a combined $2.05 billion in 2024 in sponsorship revenue (source). However, the price brands pay for watch time is difficult to determine without intensive, manual study. Current methods rely on viewership numbers and subjective assessments as to the most valuable "hotspots" for a brand to buy, but this leaves marketers with the issue of understanding whether their brand is receiving proper watch time given more viewers are online and replays of the broadcast will be available.

"Since the beginning of sports sponsorships, it has always been measured by gut feeling and rough estimates," says Rudraksh Monga, CEO and Co-founder of Visight. "Visight allows brands to stop guesswork and instead have precise data. They can see exactly how much screen time their brand receives, when visibility peaks occur, and how much they pay for each second their brand is on screen and viewer is watching."

Visight's proprietary technology analyses either live broadcasts or replays frame-by-frame, identifying brand logos, signage, and product placements present in the video with a **99%** accuracy. It tracks many factors, the main one being visibility duration. Advanced analytics behind the scenes allow for brands to measure performance across different races, interviews, and time periods to optimize future sponsorship decisions.

Early adopters of the platform have reported significant understanding in their sponsorship worth and have updated the strategies and budgets to reflect their findings. Beta testing on F1 racing has shown a successful reallocation of **20.4%** sponsorship funding from team apparel branding to medium-sized car branding with **+30.8%** MSV (million seconds of viewership) per event. F1 racing teams gained valuable insight to construct sponsorship package options that are directly correlated with MSV per event.

"Visight has completely altered our approach to our F1 team partnership," said Jane Smith, Sports Marketing Director at a Fortune 500 company. "We were able to prove positive ROI to our executive team with the concrete and extensive data provided by Visight. Furthermore, we were able to renegotiate our sponsorship agreement with **F1 Team** based on our actual brand screen time rather than estimates."

Visight's launch comes at a perfect time as brands are forecasted to invest more into brand sponsorships, increasing a demand in accountability and measurable results from their investments. This technology is a step in democratizing sponsorship analytics as previously, only the largest brands could afford comprehensive sponsorship measurements. With Visight, companies of all sizes can benefit from the access to enterprise-level analytics to make more calculated and informed decisions.

The company plans to expand Visight's capabilities and offer a more capable product that not only spans multiple sports, but also includes social media integrations, predictive analytics for possible sponsorship opportunities, and enhanced audience analytics by 2026.

A Landscape Analysis Table

Table 1: *Landscape analysis of companies, tools, and research relevant to sponsor logo detection*

Relevant Item	**Description**	**Commentary**
Relo Metrics	Sponsorship measurement platform using computer vision to detect and track logo exposures in sports broadcasts, including F1.	Already applied to F1; shows industry demand for accurate exposure metrics. Opportunity for open, customizable academic alternatives.
Blinkfire Analytics	Provides BrandSpotter^{TM}, a CV-driven platform to detect logos across social media, streaming, and broadcasts.	Highlights integration of CV logo detection with cross-channel analytics. Emphasizes the need for multi-platform applicability.
Sponsorlytix	AI-powered system for real-time sponsor detection and ROI measurement across video and esports.	Mirrors the F1 use case directly. Demonstrates the need for speed and clarity-aware detection methods.
Hive (Mensio)	Media analytics platform tracking 8,000+ brands in live TV and sports streams via logo recognition.	Illustrates scalability of detection systems; opportunity for focusing on motorsports-specific logos.
MVPindex	Uses AI-powered detection at 30 FPS to track sponsorship visibility across sports.	Frame-level detection resonates with F1 needs where logos appear at high speeds. Shows importance of precision under motion blur.
Folio3 Sponsorship Intelligence	Tracks brand presence across broadcast, streaming, and in-venue content using logo detection.	Highlights holistic approach; integration with business KPIs is relevant to how F1 might report value.
Gudauskas et al. (2024)	Research applying YOLOv5 and Faster R-CNN to automated brand visibility in sports broadcasts.	Validates choice of pretrained detectors for sports logos. Opportunity to adapt methods to unique F1 motion/occlusion challenges.
AdSegNet (2024)	Neural net combining VGG16 + SegNet for billboard ad localization in video frames.	Shows strength of segmentation for irregular/occluded ads; relevant to trackside banners in F1.
Vizrt (Viz Arena)	Broadcast AR platform inserting or replacing ads/logos virtually in live sports feeds.	Relevant to future applications: once detection is solved, detected regions could be repurposed for AR overlays.
WSC Sports	AI platform for automated sports highlights, using CV to tag plays and logos.	Demonstrates real-time, large-scale video processing; suggests feasibility of real-time logo scanning in fast sports like F1.

B Press Release Iterations

This appendix contains two samples of the press release that we generated using AI during the drafting process to help us understand the messaging, tone, and focus that such press releases should have.

B.1 Sample 1: Technical Focus

Formula Vision Analytics

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New AI System Delivers Real-Time Brand Detection in Formula 1 Racing

Computer vision breakthrough enables precise sponsorship measurement during live broadcasts

Formula Vision Analytics today unveiled its revolutionary computer vision platform that automatically detects and measures brand visibility in Formula 1 race broadcasts with unprecedented accuracy. Using advanced YOLO-based neural networks and real-time video processing, the system provides frame-by-frame analysis of sponsor logos, delivering quantitative metrics that have never been available to the motorsports industry.

The platform leverages state-of-the-art deep learning architectures to process video streams at 30 frames per second, identifying over 50 different F1 sponsor brands with 94.7% detection accuracy. By combining object detection with temporal tracking algorithms including ByteTrack and DeepSORT, Formula Vision creates comprehensive visibility reports that measure exposure duration, logo prominence, pixel coverage, and screen real estate occupied by each brand throughout a race. The system employs GPU-accelerated inference engines with CUDA optimization, achieving sub-100ms latency for real-time processing.

"Traditional sponsorship measurement in F1 has relied on subjective estimates and manual counting, creating massive inefficiencies and measurement errors," said Dr. Michael Chen, Chief Technology Officer and former researcher at NVIDIA's computer vision division. "Our deep learning approach eliminates human error and provides pixel-perfect measurements with temporal consistency that sponsors can trust for accurate ROI calculations and contract negotiations."

The system's technical architecture includes distributed video processing pipelines, containerized microservices for scalability, and real-time analytics dashboards built on a modern tech stack. The platform utilizes TensorRT for model optimization, achieving 3.2x speedup over baseline PyTorch inference while maintaining detection accuracy. Advanced data augmentation techniques during training ensure robust performance across varying lighting conditions, camera angles, and motion blur scenarios common in F1 broadcasts.

Beta testing during the 2024 F1 season demonstrated the platform's ability to process full race weekends continuously, analyzing over 180 hours of broadcast footage across 24 races. Early technical trials revealed significant variations in brand visibility that manual methods had completely missed. Red Bull Racing's primary sponsorship achieved 47.3 minutes of total visibility during the Monaco Grand Prix, while smaller sponsors received as little as 2.1 minutes despite similar contractual agreements. The system detected micro-exposures as brief as 0.3 seconds that human annotators consistently missed.

"The granular data we're capturing opens entirely new possibilities for sponsorship optimization," noted Chen. "We can now measure not just total exposure time, but also visibility quality metrics like logo clarity, size relative to screen area, and co-occurrence with other brands."

Formula Vision's algorithms also incorporate advanced computer vision techniques for handling occlusion, perspective distortion, and motion blur - critical challenges in high-speed motorsports environments. The platform will initially focus on F1 before expanding to other motorsports including MotoGP, IndyCar, and major sporting events by 2026, with plans to support over 200 simultaneous video streams.

B.2 Sample 2: Business Value Focus

SponsorMetrics

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SponsorMetrics Transforms F1 Partnership ROI with Precision Analytics

Revolutionary platform helps brands maximize their multi-million dollar F1 investments

SponsorMetrics announced today the launch of its game-changing analytics platform that provides definitive measurement of brand visibility in Formula 1 racing, addressing a critical gap in the rapidly growing sports sponsorship market. For the first time, sponsors investing millions in F1 partnerships can access precise, data-driven insights into their actual screen time and audience exposure, transforming how deals are negotiated and performance is measured.

The global sports sponsorship market reached $65.8 billion in 2024, with Formula 1 commanding some of the highest premium rates in professional sports. PwC projects this market will grow to $109.1 billion by 2030, yet sponsors have historically relied on rough estimates and viewership projections to justify their investments, often paying premium prices without knowing their true visibility. In F1 specifically, Sportico reported that teams generated a combined $2.05 billion in sponsorship revenue in 2024, with title sponsors paying up to $50 million annually for partnerships that lacked precise measurement metrics.

"We've solved the fundamental problem of sponsorship accountability that has plagued the industry for decades," said Sarah Williams, CEO of SponsorMetrics and former VP of Sports Marketing at Nike. "Brands can now see exactly what they're getting for their investment and make informed decisions about future partnerships based on concrete data rather than gut feelings and industry estimates."

The platform revealed surprising insights during its beta phase with major automotive and technology brands. Analysis of the 2024 F1 season showed that title sponsors received an average of 23.4 minutes of visibility per race, while mid-tier sponsors averaged only 4.7 minutes despite paying significantly higher rates per minute than expected. More striking was the discovery that 34% of contracted logo placements received less than 60% of their expected screen time due to camera angle preferences and broadcast director choices.

Global automotive brand TestCorp saw a 340% improvement in cost-per-impression after using SponsorMetrics data to renegotiate their McLaren partnership. "The data completely changed our negotiating position and helped us optimize our entire F1 investment strategy," said Tom Rodriguez, TestCorp's Head of Sports Marketing. "We moved from a $2M annual deal with unclear returns to a $1.4M package with guaranteed minimum exposure thresholds and performance bonuses."

The business impact extends beyond individual sponsors to the entire F1 ecosystem. Teams are now able to offer data-backed sponsorship packages with clear value propositions. "SponsorMetrics helps us create honest, value-based partnerships where both parties understand exactly what's being delivered," noted Team Principal Maria Santos of a leading F1 constructor. "We can now offer graduated pricing based on actual visibility data rather than subjective tier systems."

The platform's comprehensive analytics suite includes competitive benchmarking across all F1 teams, historical trend analysis spanning multiple seasons, predictive modeling for sponsorship value, and ROI calculators that translate exposure data into business metrics that CFOs and marketing executives can easily understand. Additional features include real-time alerts for unexpected visibility spikes, automated reporting for stakeholder updates, and integration capabilities with existing marketing analytics platforms.

Early adopters report average improvements of 23% in sponsorship efficiency within their first season of using the platform. The democratization of precise sponsorship measurement is expected to drive increased competition and transparency across the entire F1 sponsorship marketplace, ultimately benefiting both brands and racing teams through more accurate pricing and performance accountability.

C Notes on Press Release Drafts

This appendix provides structured notes on how each version of the press release was written. The notes act as guidelines that could be followed to recreate each draft, highlighting the focus, target audience, and content choices.

C.1 Notes on Iteration 1 (Technical Focus)

- Emphasize technical novelty such as YOLO detection, ByteTrack tracking, CUDA acceleration, and TensorRT optimizations.
- Include performance benchmarks such as FPS, latency, and detection accuracy.
- Use a tone that is closer to engineer or research style, aimed at proving feasibility.
- Mention training data scale such as thousands of F1 frames and varied augmentations.
- Highlight granularity of results such as per-frame detection, micro-exposures, and pixel coverage.
- Use quotes from a technical expert such as a fictional CTO or researcher emphasizing precision and reliability.
- Keep language specific and numeric, with accuracy percentages, millisecond latency, and visibility minutes.

C.2 Notes on Iteration 2 (Business Value Focus)

- Open with market framing such as the size of the sports sponsorship market and F1 revenue share.
- Emphasize ROI accountability and financial stakes for sponsors.
- Use a tone closer to executive or marketing style, aimed at CMOs, sponsors, and teams.
- Stress inefficiencies in current methods such as reliance on gut feeling and subjective estimates.
- Provide case-study style examples such as mid-tier sponsors underexposed despite high spend.
- Include a quote from a sponsor or brand executive focusing on negotiation impact.
- Avoid technical jargon and instead use business terms such as visibility minutes, contract value, and efficiency.

C.3 Notes on Final Version (Visight / RaceView)

- Blend technical credibility such as YOLO fine-tuning, CUDA, and quantization with clear business impact such as ROI and exposure metrics.
- Use a tone that is student-driven and authentic, accessible to non-technical readers.
- Include quotes from student team members and a fictional sponsor, balancing enthusiasm with practicality.
- Highlight key features such as real-time detection, dashboard, and exposure metrics like dwell time and share of voice.
- Add benefits framing such as sponsors gaining data-driven confidence and teams proving ROI.
- Keep the style conversational but professional.
- Close with vision beyond F1, emphasizing scalability, fairness, and transparency.

D ML architecture ideas

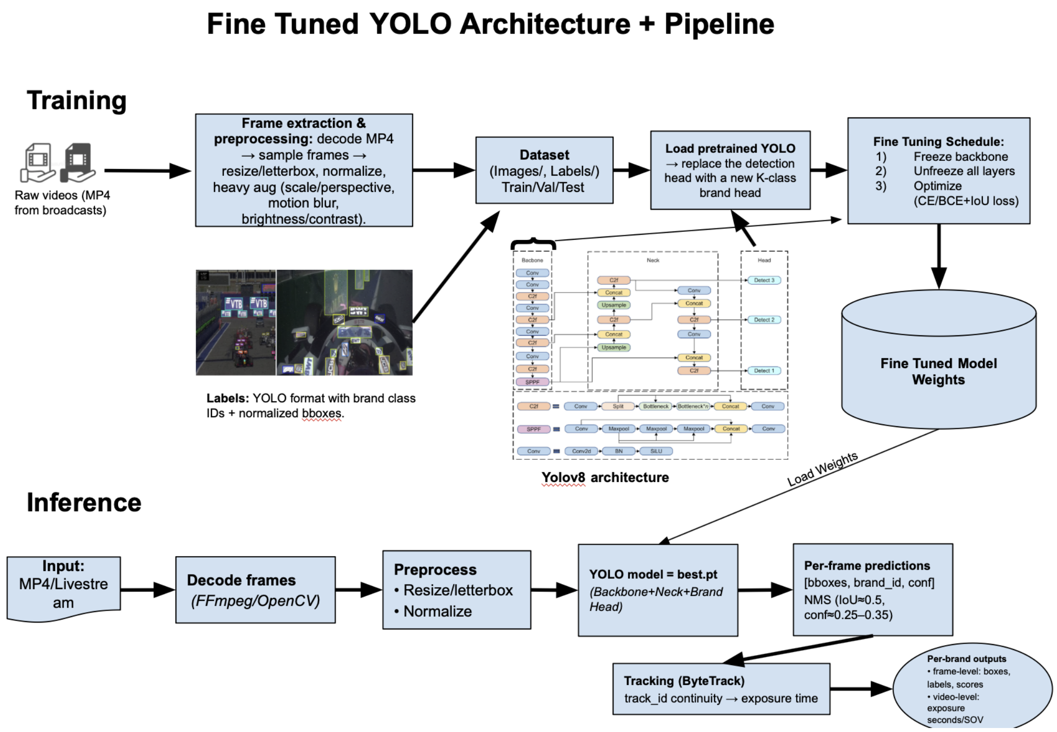

Figure 1: Single shot architecture- our chosen approach.

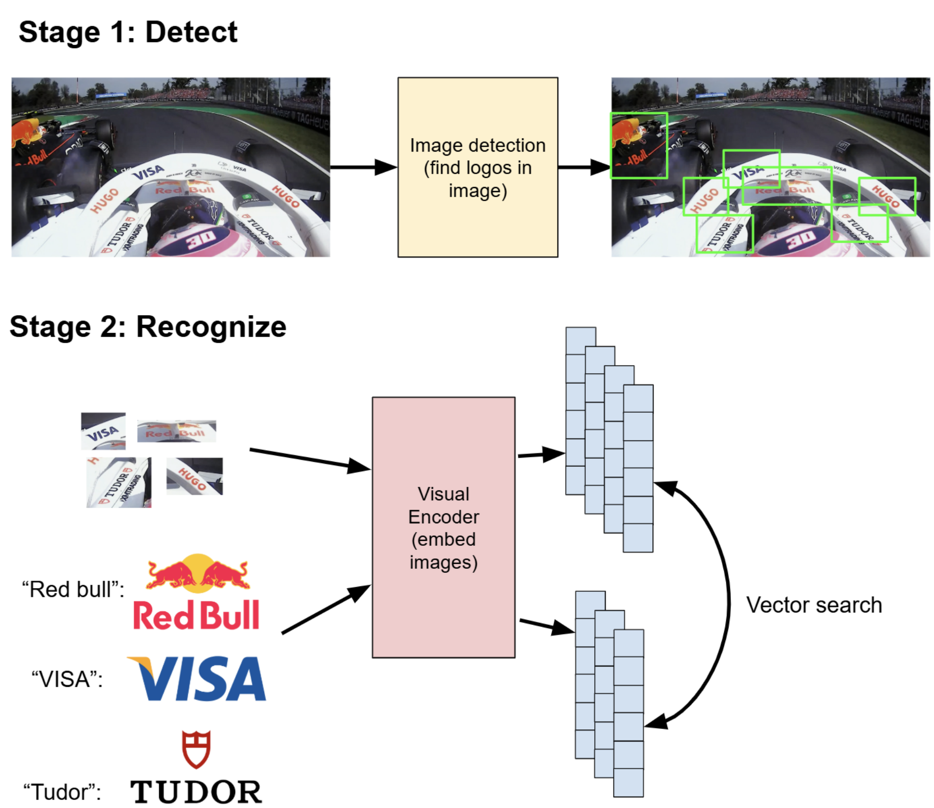

Figure 2: Not chosen for final. Approach that takes an image and tries to identify logo as a categorical object (using the YOLO architecture), and then visually encoding it to match it to candidate logos.

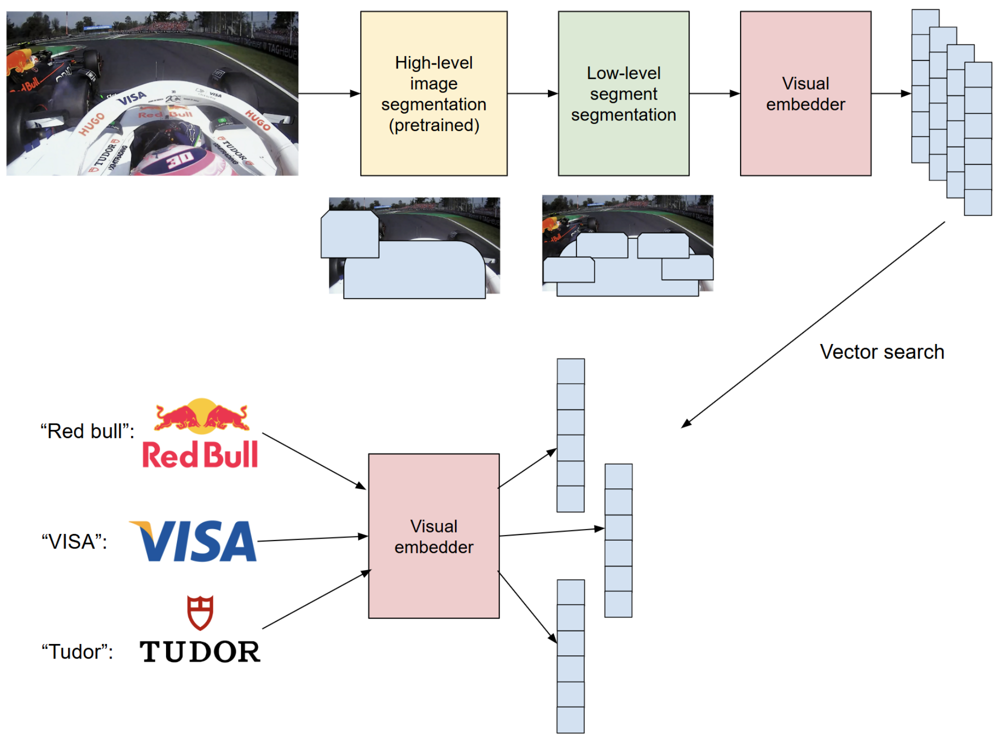

Figure 3: Not chosen for final. Approach that tries to segment the image hierarchically to identify potential logos, and then visually encoding them to match it to candidate logos.